

Accessibility by and to transit; comparison with transit Supply Index results

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Introduction

transit Supply Index (SI) has been previously developed [Currie and Senbergs, 2007],

Previous development of code to calculate SI scores from GTFS datasets as an R package [Reynolds, 2024].

The rest of this document is structured as follows:

Research context

The Supply Index

The Supply Index (SI) equation is shown in the margin figure¹, in which: (1) $SI_{area,time}$ is the Supply Index for the area of interest and a given period of time; (2) $Area_{Bn}$ is the buffer area for each stop (n) within the area of interest. In Currie and Senbergs (2007) this was based on a radius of 400 metres for bus and tram stops, and 800 metres for railway stations; (3) $Area_{area}$ is the area of the area of interest; and (4) $SL_{n,time}$ is the number of transit arrivals for each stop for a given time period.

Methodology

R [R Core Team, 2024], a widely used and readily available statistical programming language, was adopted for code development. The package development setup and workflow described by Wickham and Bryan [2023] was adopted in this study.

Various existing packages were relied upon including: the sf package [Pebesma, 2024] for geospatial analysis; the tidyverse [Wickham et al., 2019]; gtfstools [Herszenhut et al., 2025]; and tidytransit [Poletti et al., 2024]. Some code was adapted from the tidytransit, gtfstools and other package's examples, vignettes and other documentation.

$$SI_{area,time} = \sum \frac{Area_{Bn}}{Area_{area}} * SL_{n,time} \quad (1)$$

¹ Minor adjustments have been made to generalise the equation, as Currie and Senbergs (2007) focus was the context of Melbourne's Census Collection Districts (CCD) and calculations based on a week of transit service. CCDs predate the introduction of Statistical Areas 1, 2, 3, and 4 (SA1, SA2, SA3, SA4), and other geographical divisions currently used by the Australian Bureau of Statistics (ABS), which may be more familiar to readers.

Tidytransit is licensed under GPL-2 | GPL-3, while gtfstools is licensed under the more permissive MIT license. For simplicity, the gtfssupplyindex package developed here is licensed under GPL-3 so as to match tidytransit.

Cases

Code development was undertaken using a number of cases for testing purposes, being: the Mornington Peninsula Tourist Railway, and Melbourne.

The Morning Peninsula Tourist Railway runs on Sundays and Wednesdays between Moorooduc and Mornington, with an intermediate stop at Tanti Park². A GTFS feed from 2018 was selected. ABS data was also used, primarily through the strayr and absmaps-data packages [Cowgill, 2025]. The Mornington Peninsular SA3 zone and the SA1 zones contained within were adopted as the areas_of_interest.

Larger scale testing was performed using the Victorian GTFS feed, published by Public Transport Victoria (PTV), sourced via Transit Mobility Data, [2023] for historical feeds. Again, ABS data was used for the areas_of_interest.

Results

Discussion and conclusions

References

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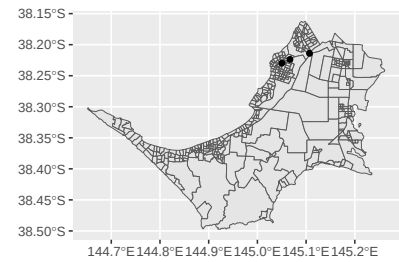


Figure 1: Mornington Peninsula SA1 zones and location of Mornington Tourist Rail stops.

²<https://transitfeeds.com/p/mornington-railway/806/latest/stops>

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